

2 — Hardware Setup & Wiring Guide

Audience: Field Technician / Electrical Engineer **Replaces:** Root-level
installation_guide.md (now deleted)

Parts Required Per Station

Item	Specification	Notes
Waveshare ESP32-S3 A7670E	4G LTE + GPS development board	Main controller
PZEM-004T v3.0	AC Power Meter module	Sensor (100A split-core CT included)
1NCE IoT SIM Card	Micro-SIM, flat-rate €10/10 years	Or any LTE SIM with APN credentials
DIN-rail PSU	5V DC / 2A minimum (e.g. Mean Well MDR-10-5)	Powers the ESP32 board
IP65 plastic enclosure	Min. 150×100×70 mm with cable glands	Houses the board inside the cabinet
4G Antenna	External SMA whip antenna	Supplied with Waveshare board
GNSS Antenna	Passive or active GPS patch antenna	Supplied with Waveshare board

Section 1 — SIM Card Activation (1NCE)

- Go to the [1NCE Customer Portal](#) and create an account.
- Click **Activate SIM Card** → enter the **ICCID** and **Activation Code** printed on the SIM plastic holder.
- Wait **10–30 minutes** for network allocation.
- Insert the SIM into the **Micro-SIM slot** on the Waveshare board. Golden contacts face **downward** toward the PCB.

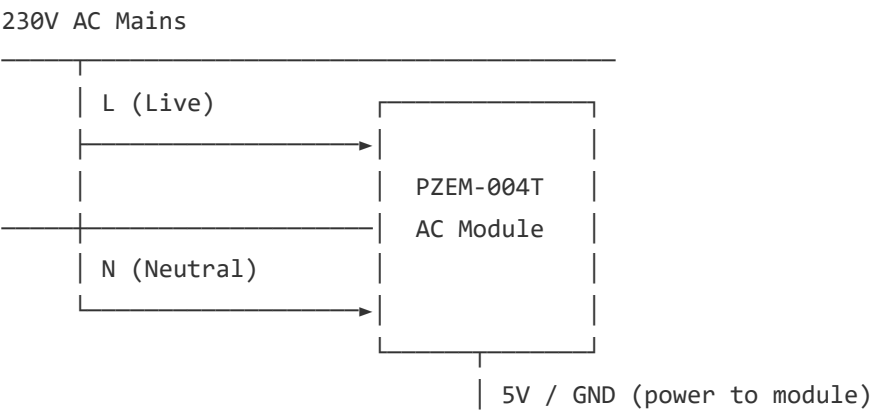
Note: The firmware is pre-configured for `iot.1nce.net` APN — no username or password required.

Section 2 — PZEM-004T Wiring

The PZEM-004T communicates with the ESP32 board using **Modbus RTU over UART** (Serial2).

2.1 Power Connections

The PZEM-004T module needs a **5V power supply** and must be connected to the **AC line** being measured:



Connect the **PZEM-004T power input terminals** (`L` and `N`) to the mains supply (or to the load/pump side — either works). The module draws ~1W from the mains for self-power.

2.2 Current Transformer (CT Clamp)

- Clip the **included split-core CT clamp** around a **single conductor** of the pump's supply cable (L1, L2, or L3 — pick one active phase).
- Do **not** clamp around both L and N together — the magnetic fields cancel and you will read zero.
- Plug the CT into the round jack socket on the PZEM module.

2.3 UART Serial Data Wiring (PZEM ↔ ESP32-S3)

Connect the PZEM-004T TTL serial header to the ESP32-S3 board:

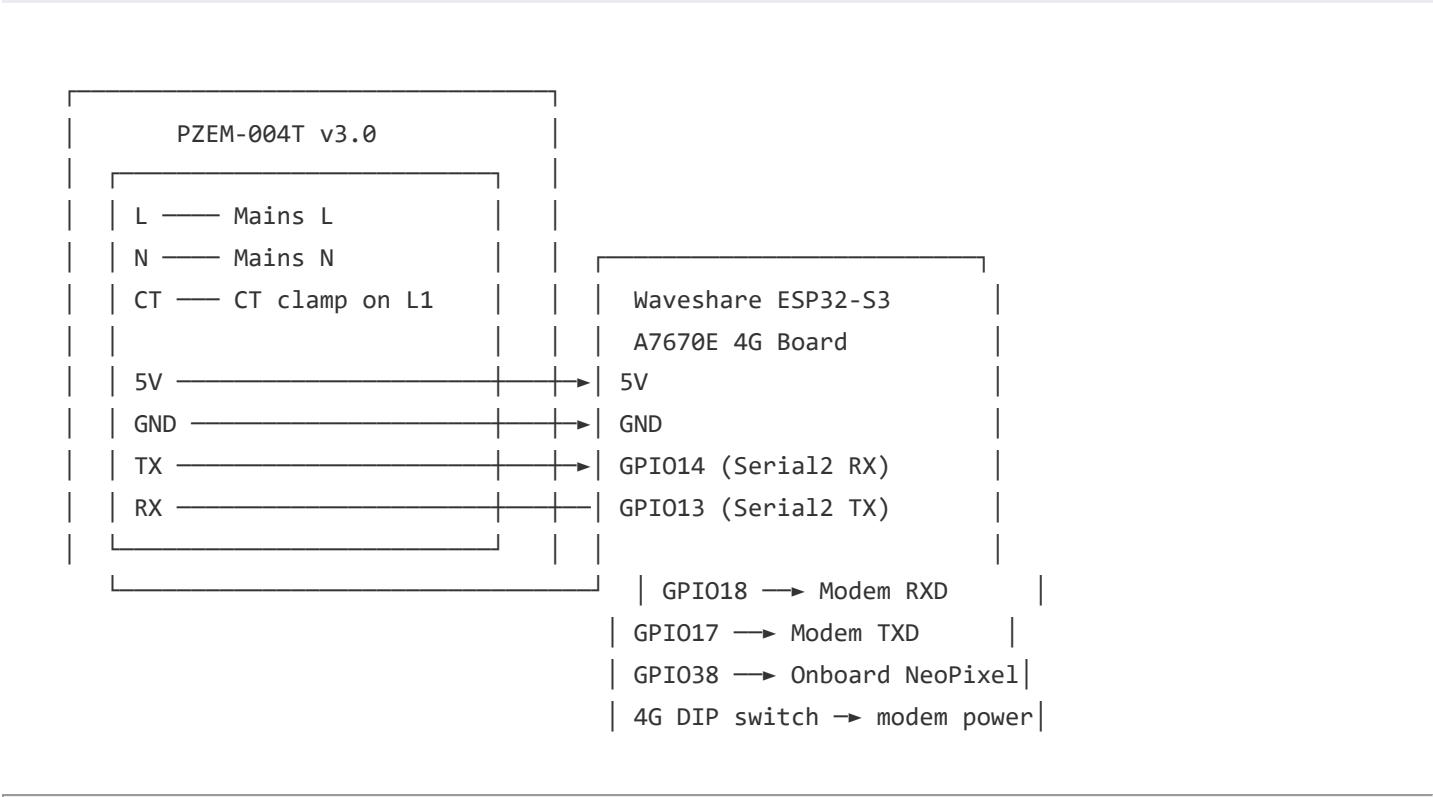
PZEM-004T Pin	ESP32-S3 GPIO	Wire Color Suggestion
5V	5V (board header)	Red
GND	GND (board header)	Black

PZEM-004T Pin	ESP32-S3 GPIO	Wire Color Suggestion
TX	GPIO 14 (Serial2 RX)	Yellow
RX	GPIO 13 (Serial2 TX)	Green

⚠️ **Important:** The PZEM TX connects to the ESP32 RX (GPIO 14), and PZEM RX connects to ESP32 TX (GPIO 13). Cross-connect TX→RX.

⚠️ **Voltage:** PZEM TTL serial operates at **3.3V logic levels** — safe for direct connection to the ESP32-S3 (also 3.3V). Do not connect to 5V-only microcontrollers without a level shifter.

2.4 Wiring Diagram (ASCII)



Section 3 — Antenna Placement

4G Cellular Antenna

- Screw the external **SMA whip antenna** onto the gold SMA connector.
- If the main electrical cabinet is **metal**, route the antenna **outside** through a rubber-gasketed hole or place it near a non-metallic vent. A metal cabinet will block LTE signals.
- In weak signal areas, use a high-gain external whip antenna (5–7 dBi).

GPS / GNSS Antenna

- Connect the **GPS patch antenna** to the U.FL/IPEX connector labeled `GNSS` on the board.

- Mount the antenna **horizontally** with a **clear view of the sky** for fastest satellite fix.
- Stick it to the top surface of the plastic enclosure, or route it outside the metal cabinet.

Section 4 — Enclosure Mounting

1. Mount the Waveshare board inside an **IP65 plastic enclosure** using M3 standoffs or adhesive PCB standoffs.
2. Pre-drill cable glands for:
 - Mains power cable (to DIN-rail PSU → 5V to board)
 - PZEM power + serial cable (PZEM is usually mounted separately near the mains entry)
 - CT clamp cable
 - Antenna cable (if routing outside)
3. Mount the enclosure inside the electrical cabinet using double-sided mounting tape or DIN-rail clips.
4. **Do not use a metal enclosure for the ESP32 board** — it will block the 4G signal.

💡 The PZEM-004T module can be mounted separately inside the cabinet near the mains terminals, with only the 4-wire UART cable running to the ESP32 board enclosure.

Section 5 — Power Supply

Supply	Specification
Mains Input	230V AC, 50Hz
DIN-Rail PSU Output	5V DC, ≥2A
Board Supply	Connect PSU 5V/GND to board's 5V header or USB-C
PZEM Power	Taken directly from mains terminals (self-powered)

Section 6 — LED Status Indicator

The onboard WS2812B NeoPixel RGB LED on GPIO 38 shows the current device state:

LED Pattern	Meaning
Slow White Pulse	Booting / Initializing
Fast Blue Blink (2 Hz)	Connecting to 4G cellular network
Fast Blue Blink (4 Hz)	Authenticating with Firebase cloud

LED Pattern	Meaning
Green Heartbeat (1 Hz)	Normal operation — fully online
Brief Cyan Flash	Uploading telemetry packet
Slow Yellow Pulse	Searching for GNSS/GPS satellite fix
Fast Orange Blink (4 Hz)	Current/Voltage reading crossed alert threshold
Slow Red Blink (1 Hz)	Soft reset (AT+CRESET) in progress
Dim Purple Pulse	Degraded Mode (modem hard-locked, offline data queueing)
White Strobe	Critical alert notification

Note: The board's I2C battery gauge uses GPIO 8 (SDA) and GPIO 9 (SCL), so the status LED must not share those pins.

Next: [Firmware & Provisioning](#) →